



Project Title

LeafDoc

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Date :

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Introduction

LeafDoc, an AI-assisted system designed for plant identification and disease diagnosis using image-based analysis. Identifying plant species and detecting plant diseases at an early stage is often challenging for farmers and non-expert users due to dependence on expert knowledge and traditional manual methods.

LeafDoc addresses this challenge by providing a web-based solution that first identifies the plant from an uploaded image and then analyzes it for possible diseases. Based on the analysis, the system provides basic treatment suggestions and plant care guidance, aiming to support farmers, gardeners, and students through an accessible and easy-to-use plant health monitoring tool.



Objectives

The primary objectives of the LeafDoc project are as follows:

1 Plant Identification

To identify plant species from user-uploaded images using AI-assisted image analysis.

3 Recommendation and Guidance

To provide basic treatment suggestions and plant care recommendations for identified diseases.

2 Plant Disease Detection

To analyze identified plants for possible diseases based on visual symptoms.

4 User Assistance and Accessibility

To develop a user-friendly web-based system with chatbot support for assisting non-expert users.

Requirement Analysis

Software Requirements

- Frontend: HTML, CSS, JavaScript
- Backend: Node.js
- Tool: Visual Studio Code

Hardware Requirements

- Processor: Dual-core or higher
- RAM: Minimum 4 GB
- Device: Laptop
- Internet: Required

Proposed System Modules

The LeafDoc system follows a structured workflow consisting of the following modules:



Image Input Module

Users upload an image of a plant or leaf through the web-based interface for analysis.



Plant Identification Module

The uploaded image is analyzed using a generative vision-language AI service to identify the plant species.



Disease Analysis Module

Based on the identified plant, the system analyzes visual symptoms to detect possible plant diseases.



Recommendation & User Interaction Module

The system provides basic treatment suggestions and plant care guidance, along with chatbot-based user assistance.

LeafDoc Modules

Home Page:

The home page provides an overview of the LeafDoc system, highlighting its purpose of plant identification and disease diagnosis using AI. It offers a clean and user-friendly interface with navigation options for accessing different features of the application.

 LeafDoc

HomeFeaturesHow It WorksAboutGet Started

Discover the World of Plants

LeafDoc helps you identify plants, diagnose diseases, and learn how to care for your green companions with cutting-edge AI.

Get Started Free

 LeafDoc

HomeFeaturesHow It WorksAboutGet Started

Why Choose LeafDoc?

LeafDoc combines advanced AI with a user-friendly interface to make plant care simple and effective for everyone.



Instant Plant ID

Snap a photo, and our AI will identify the plant species with remarkable accuracy. Say goodbye to guesswork!



Disease Detection

Upload an image of a symptomatic plant. LeafDoc's AI analyzes it to detect potential diseases and offers insights.

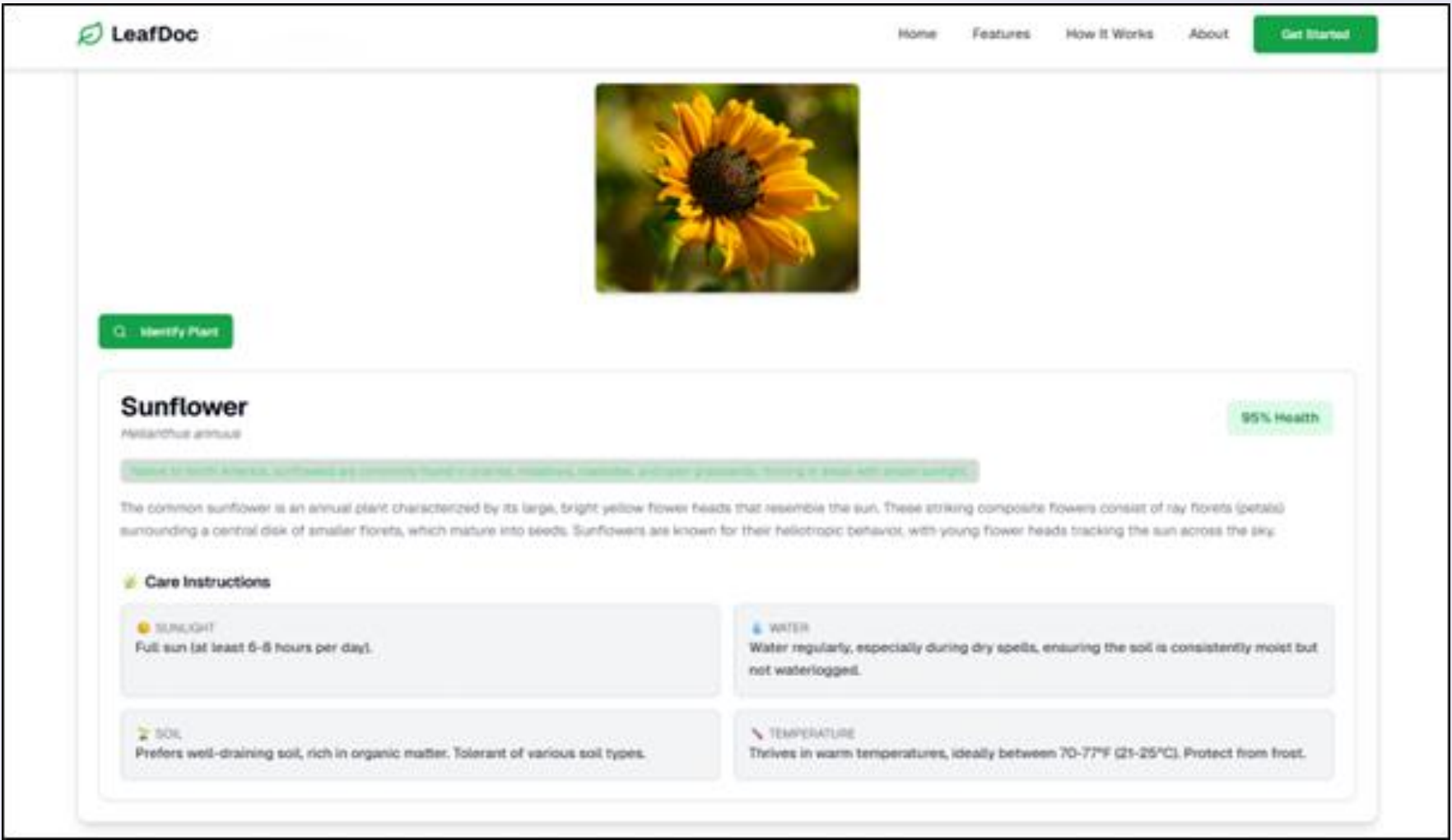
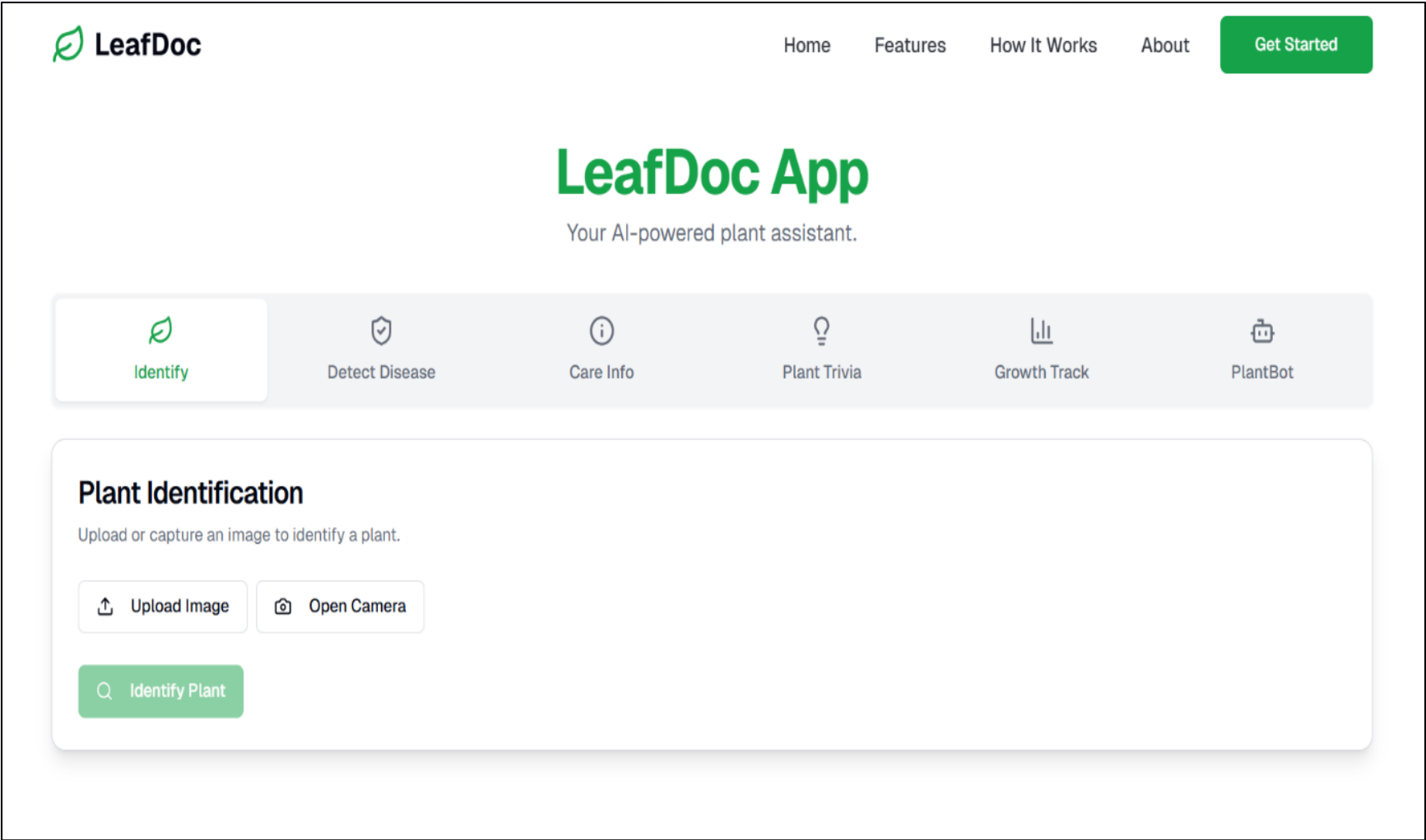


Expert Care Guides

Access concise, easy-to-follow care information for your identified plants, covering watering, sunlight, and more.

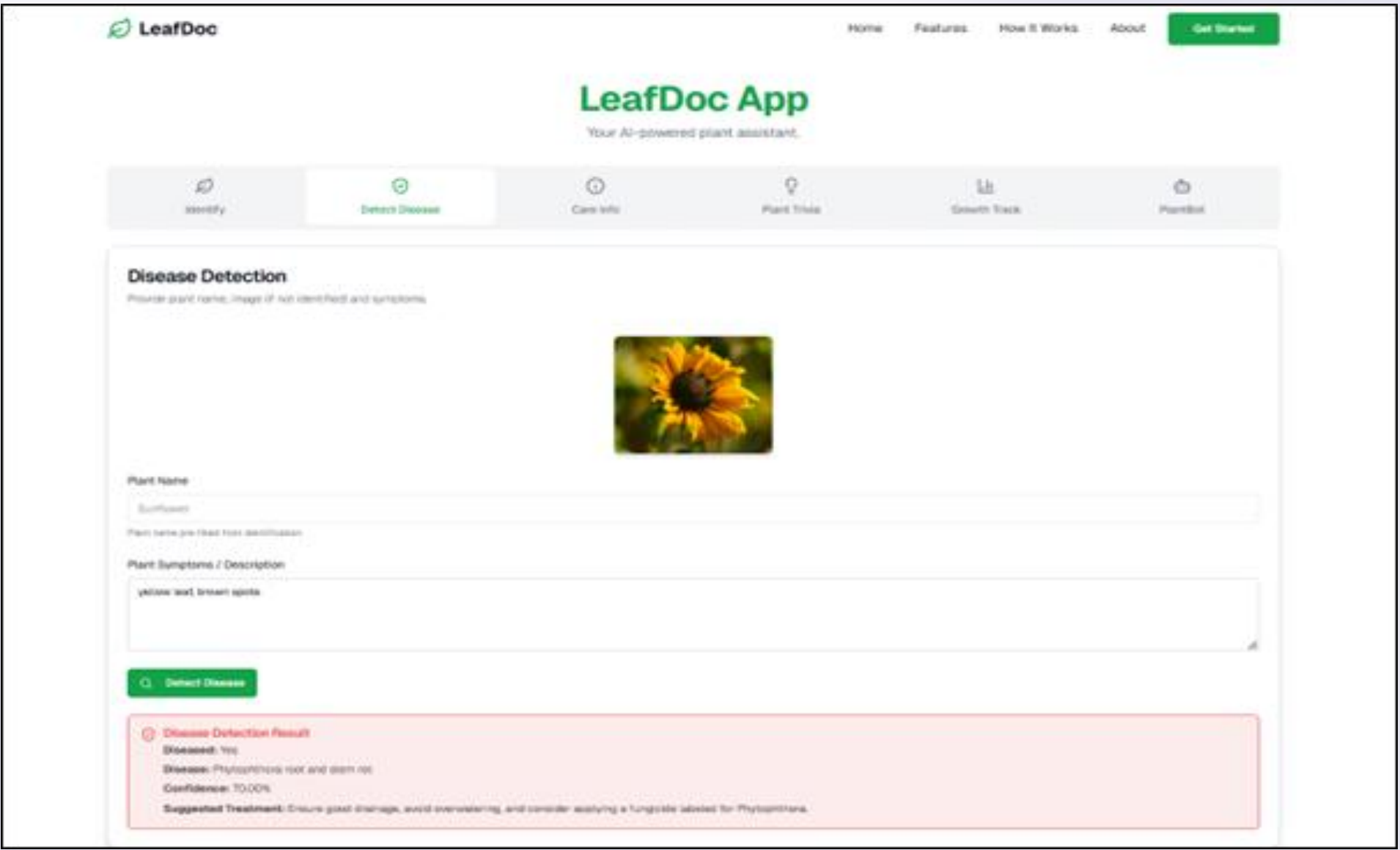
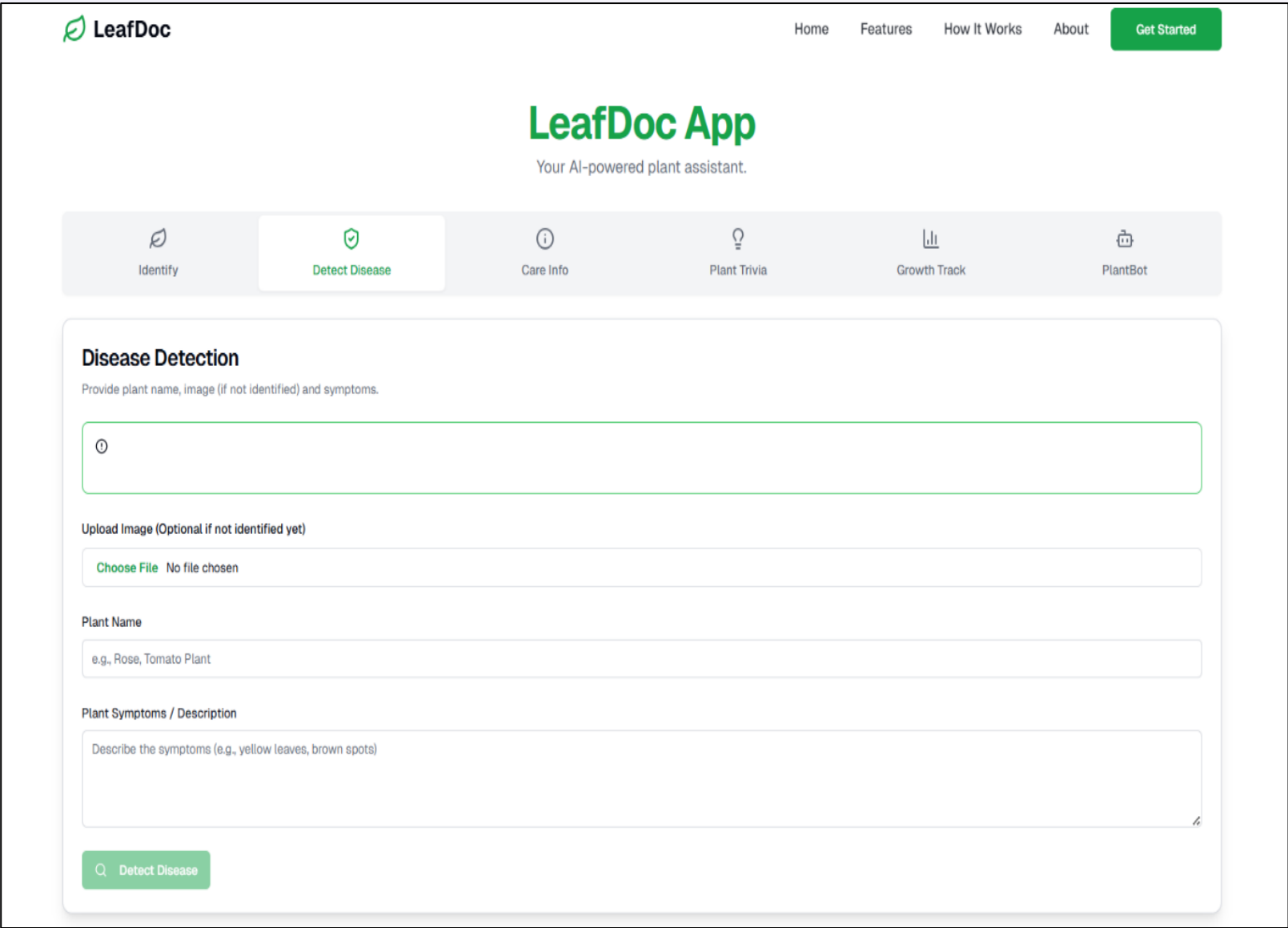
Plant Identification Module:

This module allows users to upload or capture plant images for identification. The AI model analyzes visual features and accurately determines the plant species along with relevant details.



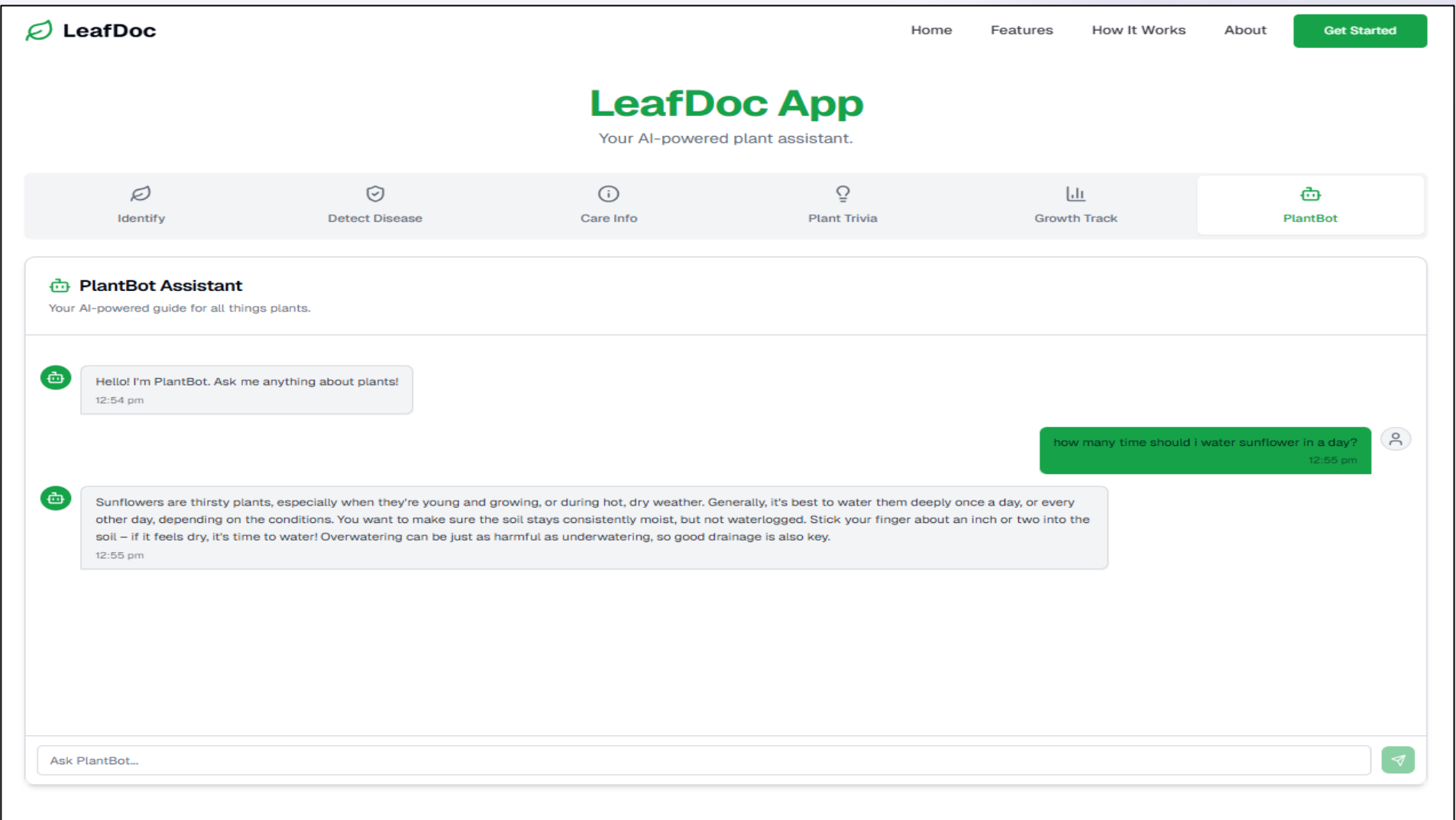
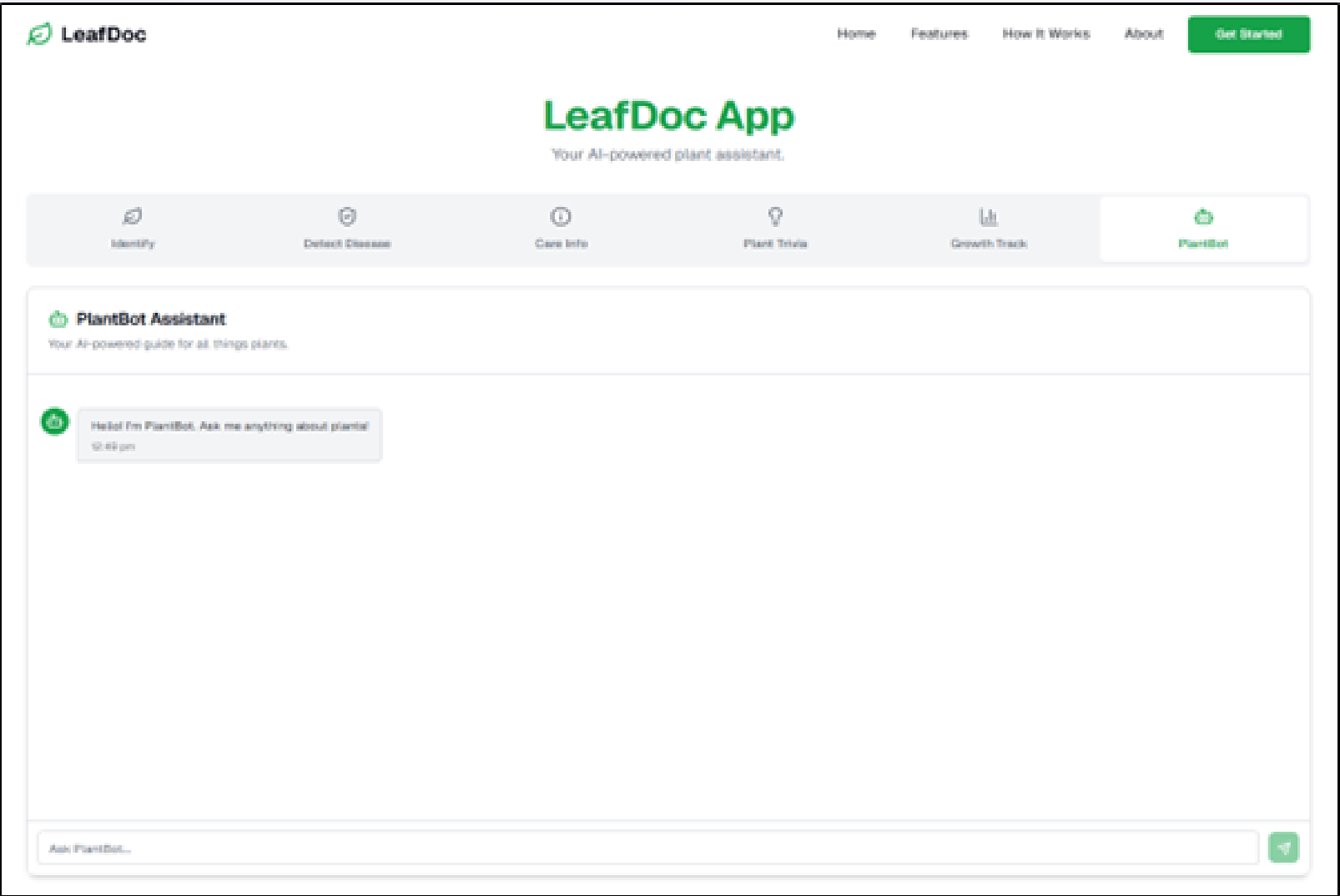
Disease Detection Module:

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PlantBot (Chatbot Module):

The chatbot module provides interactive assistance by answering plant-related queries. It offers real-time guidance on plant care, diseases, and general information using AI.



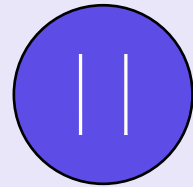
Advantages

- Easy to use for beginners
- Fast and automated results
- Combines identification and disease detection
- Provides treatment suggestions
- Reduces dependency on experts

Disadvantages

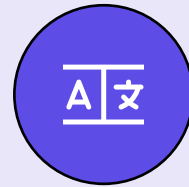
- Accuracy depends on image quality
- Limited detection for rare plants
- Requires internet connection

Future Scope



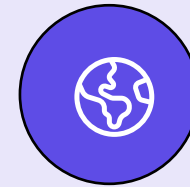
Development of a Mobile Application

Extend the current web-based system into a mobile application for improved accessibility and convenience.



Multi-Language Support for Regional Users

Provide support for multiple regional languages to make the system accessible to a wider user base.



Integration of Environmental Data

Incorporate environmental factors such as climate and location to enhance plant care and recommendation accuracy.

Conclusion

LeafDoc provides an efficient and user-friendly solution for plant identification and disease detection using artificial intelligence.

The system successfully integrates AI with web technologies to analyze plant images and generate accurate results along with treatment suggestions and care guidance. It reduces dependency on manual methods and expert knowledge, making plant health monitoring more accessible to beginners and non-expert users.

Overall, the system demonstrates practical applicability in agriculture, gardening, and educational use, while also offering flexibility for future enhancements.

Thank You !